



**LAPORAN PROGRAM
BEASISWA RETOOLING KOMPETENSI VOKASI
DOSEN PENDIDIKAN TINGGI VOKASI**

TAHUN ANGGARAN 2017

Oleh:

SETYO NUGROHO

POLITEKNIK ELEKTRONIKA NEGERI SURABAYA

**DIREKTORAT PENGEMBANGAN KELEMBAGAAN PERGURUAN TINGGI
DIREKTORAT JENDERAL KELEMBAGAAN, ILMU PENGETAHUAN,
TEKNOLOGI, DAN PENDIDIKAN TINGGI**

**KEMENTERIAN RISET, TEKNOLOGI DAN PENDIDIKAN TINGGI
TAHUN 2017**

KATA PENGANTAR

Segala puji dan syukur kepada allah swt yang telah memberikan kesempatan untuk belajar dan mengembangkan diri melalui program retooling dosen vokasi kemristekdikti

Semoga dengan adanya program ini dapat membuka wawasan baru bagi para dosen tentang penerapan ilmu pengetahuan dan teknologi serta bisa mendapatkan pengalaman industry yang bisa diterapkan kedalam system belajar mengajar di perguruan tinggi masing-masing

Penyusun

Setyo nugroho

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BAB I PENDAHULUAN

A. Latar Belakang Kegiatan

Politeknik Elektronika Negeri Surabaya (PENS) adalah salah satu perguruan tinggi yang unggul dalam bidang elektronika. PENS telah banyak menghasilkan lulusan yang ahli di bidang robotika, otomasi industri, elektronika, kelistrikan dan distribusi, dan lain sebagainya.

Sejak tahun 2016 PENS telah menjalin kerjasama dengan PT. Garuda Maintenance Facilities Airasia (PT. GMF Aeroasia) untuk mendidik mahasiswa dengan program studi D3 Electrical Avionic (D3 EA). Lulusan dari program ini telah diikat oleh PT. GMF untuk menjadi karyawan dengan telah memiliki lisensi resmi dari otoritas penerbangan di Indonesia

Program D3 EA ini mempelajari secara khusus tentang system kelistrikan dan perangkat elektronika yang ada pada pesawat terbang. Diharapkan lulusan program ini dapat melakukan perbaikan, perawatan, dan menyelesaikan masalah yang timbul pada system kelistrikan dan instrument pesawat terbang

Untuk dapat menghasilkan lulusan yang berkompeten perlu ditunjang oleh tenaga pengajar yang berkompeten pula. Untuk pengajar program D3 EA disyaratkan telah memiliki lisensi untuk dapat mengajar yang dikeluarkan oleh otoritas penerbangan Indonesia. Untuk mendapatkan lisensi ini bisa dengan cara mengikuti pelatihan dan magang industri yang materinya telah ditentukan oleh otoritas.

Dengan alasan itu maka program retooling ini akan sangat mendukung pengembangan diri dosen dalam upaya pemenuhan syarat agar dapat mengajar di bidang aviasi yang mana membutuhkan lisensi khusus berdasarkan pengetahuan dan pengalaman di bidang aviasi

B. Tujuan Kegiatan

Dari latar belakang tersebut dapat dibuat tujuan dari program ini adalah sebagai berikut:

1. Memberikan pengetahuan di bidang aviasi khususnya perawatan pesawat terbang

2. Memberikan keahlian dan kompetensi yang bersertifikasi di bidang perawatan pesawat terbang sebagai syarat untuk dapat mengajar di bidang aviasi

C. Manfaat Kegiatan

Setelah mengikuti kegiatan ini diharapkan dosen bisa memberikan system dan materi pengajaran yang sesuai dengan kebutuhan industry perawatan pesawat terbang, serta juga dapat membangun mental kerja dan profesionalitas yang dibutuhkan seorang teknisi pesawat terbang.

BAB II RENCANA KEGIATAN

A. Tempat Kegiatan

Kegiatan Retooling ini berlangsung di dua tempat dengan topik yang berbeda

1. Didactic dan Learning Theory diselenggarakan di ZbW St Gallen, Swiss
2. Practical and Maintenance Practice diselenggarakan di SR Technic, Zurich

B. Waktu Kegiatan

Kegiatan berlangsung selama empat puluh satu hari dari tanggal 12 November 2017 sampai dengan 22 Desember 2017

Kegiatan dilakukan secara bergantian antara di ZbW St Gallen dan SR Technic

C. Lingkup Kegiatan

Kegiatan meliputi teori tentang pembelajaran serta metode pengajaran teori, serta metode pembelajaran untuk praktikal yang berkenaan dengan perawatan pesawat terbang

D. Deskripsi Rinci Rencana Kegiatan

Kegiatan dilakukan selama 12 hari di SR Technic Zurich dan 17 hari di ZbW St Gallen.

Di SR Technic akan mempelajari tentang pembelajaran Maintenance Practice meliputi metode pembelajaran, safety precaution, praktek, observasi lapangan. Topik yang dipelajari meliputi tentang avionics dan mekanik.

Di ZbW akan mempelajari cara merancang sampai menyampaikan pembelajaran. Kegiatan akan dimulai dengan cara mendapatkan informasi tentang sumber daya pembelajaran yang meliputi para siswa, lingkungan belajar, fasilitas penunjang, bahan ajar. Kemudian akan dikembangkan menjadi sebuah perencanaan pembelajaran. Berbagai macam metode pembelajaran juga akan dipelajari untuk mendukung proses belajar. Metode-metode ini akan menitikberatkan kepada kegiatan kolektif siswa. Lalu akan dipelajari metode dan cara untuk menguji pemahaman para siswa.

E. Deskripsi Luaran (output) Kegiatan

Luaran kegiatan retooling perawatan pesawat terbang ini sertifikasi TOT untuk pembelajaran Adult learning dengan judul Certificate SFAL lecture. Dan Sertifikat TOT untuk praktek bidang perawatan pesawat terbang dari SR Technics untuk bidang B1 (mechanical) dan B2 (Avionics).

PENS sedang menjalin kerjasama ikatan dinas dengan PT. GMF Aeroasia untuk menjalankan program Basic Aircraft Maintenance dengan keahlian Electronic Aviation. Output dari program ini adalah tenaga kerja yang bersertifikasi untuk memperbaiki dan merawat system kelistrikan dan instrument pesawat terbang. Untuk mengajar di program ini diwajibkan memiliki sertifikasi pengajar yang dikeluarkan otoritas penerbangan Indonesia

Di tahun-tahun berikutnya PENS akan tetap membuka program D3 EA untuk mencukupi kebutuhan tenaga kerja sebagai teknisi pesawat terbang. Kebutuhan tenaga kerja di bidang ini sudah sangat mendesak dan massif seiring dengan pertumbuhan penumpang dan pesawat terbang di Indonesia.

F. Deskripsi Dampak (outcome) Kegiatan

Setelah mengikuti program ini, akan di implementasikan rancangan kurikulum yang sesuai dengan kebutuhan industri perawatan terbang khususnya di bidang Electrical Aviation. Kurikulum ini akan mencakup teori, praktek, dan magang industry.

Sikap mental dan profesionalitas juga akan ditempa yang dituangkan kedalam kurikulum yang tujuannya adalah untuk menghasilkan tenaga kerja yang memiliki etos kerja, dan mental teknisi pesawat terbang yang dituntut memiliki integritas dan standar kerja yang tinggi.

BAB III PELAKSANAAN KEGIATAN

A. Tahap Persiapan Kegiatan

Persiapan dilakukan selama kurang lebih satu bulan yang meliputi persiapan administrative untuk ijin ke luar negeri ke Setneg dan dokumen untuk pengurusan Visa ke Swiss

B. Tahap Pelaksanaan Kegiatan

Perincian jadwal kegiatan ini bisa dilihat pada table di bawah ini;

Day	Time	ZbW St. Gallen	Gr 1: SR-Technics Kloten	Gr 2: SR-Technics Kloten
KW46				
MON, Nov 13th	14:00 – 16.30 Room 321	Welcome, Introduction (Mr. Andreas Schubiger, Mr. Gregor Thurnherr, Mr. Harry Graschi)		
TUES, Nov 14th	08:30 – 16.30 Room 321	Understanding learning and teaching (Mr. Harry Graschi, Mrs. Siham Afra)		
WED, Nov 15th	08:30 – 16.30 Room 321	Planning learning processes (SA/GR)		
THUR, Nov 16th	08:30 – 18.00 Room 222	Planning learning processes / Understanding teaching and learning/Process evaluation(GR/SA)		
FRID, Nov 17th	08:10 – 11.00 St. Gallen	At main station St. Gallen 08.10_Visit the practice of swissmechanic : Oberstrasse 153, 9000 St. Gallen, 071 274 00 30 (GR) Paradies Moschee Xhamia Xheneti, Paradiesstrasse 2, 9000 St. Gallen (20 min from swissmechanics)		
KW47				
MON, Nov 20th	08:30 – 16.30 Room 321	Designing learning enviroments: Advance Organizer, Cognitive Apprenticeship (SA/GR)		
TUES, Nov 21st	08:30 – 18.00 Room 321	Designing learning enviroments / WELL Process evaluation (TH/GR/SA)		
WED, Nov 22nd	08:30 – 16.30		Welcome, Introduction (Mr. Thomas Rissi) M7 Training Manual Theory (electrical/avionic portion) / Online CD manuals access	Welcome, Introduction (Mr. Peter Hänggi) M7 Training Manual Theory (mechanical portion) / Online CD manuals access
THUR, Nov 23rd	08:30 – 16.30		Maintenance Practices (electrical/avionic side) / Display material used (electrical/avionic side)	Maintenance Practices (mechanical side) / Display material used (mechanical side)
FRID, Nov 24th	08:30 – 16.30		Display material used (electrical/avionic side)/ Preparation as a Practical Trainer for an upcoming class	Maintenance Practices (mechanical side) / Preparation as a Practical Trainer for an upcoming class
KW48				
MON, Nov 27th	08:30 – 16.30 Room 321	Guiding group processes (GR/TH)		
TUES, Nov 28th	08:30 – 16.30 Room 321	Presenting learning contents (Mrs. Sonja Bänziger/ Mr. Gregor Thurnherr)		
WED, Nov 29th	08:30 – 18.00 Room 321	Presenting learning contents (SB/TH) & Feedbacks / Process evaluation (TH/SB)		
THUR, Nov 30th	08:30 – 16.30		M7 Maintenance Practices (electrical/avionic side) / Tools, Equipment, Infrastructure (electrical/avionic side)	Mr. Ernst Lutz: M7 Maintenance Practices (mechanical side) / Tools, Equipment, Infrastructure (mechanical side)
FRID, Dec 1st	08:30 – 16.30		Assessment as a Practical Trainer (electrical/avionic side) & Feedback	Maintenance Practices (mechanical side) / Preparation as an Assessor for an upc. class
KW49				
MON, Dec 4th	08:30 – 16.30	Enabling self-controlled learning (GR/TH)		

TUES, Dec 5th	08:30 – 16.30 Room 321	Enabling self-controlled learning (GR/TH) Assessing learning performance (TH/GR)		
WED, Dec 6th	08:30 – 18:00 Room 321	Assessing learning performance (TH/GR) Process evaluation (TH/GR)		
THUR, Dec 7th	08:30 – 16.30		Mr. Ernst Lutz : Maintenance Practices (mechanical side) / M7 Training Manual Theory / Online CD manuals access (mechanical portion)	Mr. Ruedi Scherer : M7 Maintenance Practices (electrical/ avionic side) / Training Manual Theory / Online CD manuals (electrical/avionic portion)
FRID, Dec 8th	08:30 – 16:30		M7 Maintenance Practices (mechanical side) / Display material used (mechanical side)	Maintenance Practices (electrical/avionic side) / Display material used (electrical/ avionic side)
KW50				
MON, Dec 11th	08:30 – 16.30		Maintenance Practices (mechanical side)/ Preparation as an Assessor for an upcoming class	Module 7 Maintenance Practices (electrical/avionic side) / Tools, Equipment, Infrastructure (electrical/avionic side)
TUES, Dec 12th	08:30 – 16.30		7 Maintenance Practices (mechanical side) / Tools, Equipment, Infrastructure (mechanical side)	Assessment as a Practical Trainer (electrical/avionic side) & Feedback
WED, Dec 13th	08:30 – 16.30		Maintenance Practices (mechanical side) / Soft-Skills of a Practical Trainer and Assessor	Mr. Peter Hänggi : Maintenance Practices (mechanical side) / Soft-Skills of a Practical Trainer and Assessor
THUR, Dec 14th	08:30 – 16.30		Maintenance Practices (mechanical side) / Preparation of a practical training (self-chosen topic)	Maintenance Practices (mechanical side) / Preparation of a practical training (self-chosen topic)
FRID, Dec 15th	08:30 – 16.30		Assessment as a Practical Trainer (mechanical side) / Feedback / Signing Syllabus	Assessment as a Practical Trainer (mechanical side) / Feedback / Signing Syllabus
KW51				
MON, Dec 18th	08:30 – 16.30	Proof of competences: Practice lesson (GR/SA)		
TUES, Dec 19th	08:30 – 16.30	Proof of competences: Practice lesson (GR/TH)		
WED, Dec 20th	08:30 – 18:00 Room 321	Proof of competences: Practice lesson (GR/TH) Process evaluation & Informations		
THUR, Dec 21st	08:30 – 16.30	Visit of the practice in Appenzell (TH)		
FRID, Dec 22nd	08:30 – 16.30 Room 308	Course evaluation / certification celebration (AS/TH/GR)		

Di ZbW akan mempelajari perencanaan pembelajaran, pembuatan materi dan timetable, pemilihan metode yang tepat, pemberian asesmen kepada pelajar. Kegiatan dilakukan secara teori, praktek, simulasi, kerja kelompok

Di SR Technic mempelajari metode pengajaran praktikal tentang aircraft maintenance bidang Maintenance Practice. Topik pembelajaran meliputi teknik perawatan dasar seperti wire locking, electrical wiring, inspeksi pesawat harian, pengetahuan dokumen kerja.

C. Tahap Akhir Kegiatan

Pada akhir kegiatan akan dilakukan asesmen penyampaian pembelajaran teori di ZbW dan penyampaian pembelajaran praktek di SR Tech. Asesmen meliputi tahap perencanaan pembelajaran, pembuatan materi dan timetable, pemilihan metode yang tepat, pemberian asesmen kepada pelajar. Setelah kegiatan dilakukan akan disusun laporan akhir ke Dikti

BAB IV PENUTUP

Tumbuhnya industri transportasi udara secara pesat di Indonesia harus diimbangi oleh kecukupan tenaga kerja yang merawat pesawat terbang. Perawatan pesawat terbang memiliki peran yang sangat vital dalam menjaga keamanan dan keselamatan penerbangan. Perawatan yang baik akan meminimalkan resiko kecelakaan dan insiden, mengurangi biaya operasional perusahaan penerbangan, dan menunjang kelancaran transportasi udara

A. Kesimpulan

Metode yang tepat perlu diterapkan untuk setiap topik pembelajaran. Metode yang tepat dapat dipilih dengan menyesuaikan sumber daya pembelajaran, fasilitas penunjang, dan hasil pembelajaran yang diharapkan. Dengan metode yang tepat diharapkan akan meningkatkan keterserapan materi dan keterampilan yang ingin dikuasai

B. Saran

1. Lebih banyak hands on experience di bidang perawatan pesawat
2. Lebih banyak observasi lapangan terkait pekerjaan sebagai mekanik pesawat
3. Ada peatihan terkait pesrisapan dokumen untuk program perawatan pesawat.

LAMPIRAN

1. Surat-surat ijin yang mendukung keberangkatan melaksanakan kegiatan Program Retooling 2017



**KEMENTERIAN RISET, TEKNOLOGI, DAN PENDIDIKAN TINGGI
POLITEKNIK ELEKTRONIKA NEGERI SURABAYA**

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Laman: <http://www.pens.ac.id> Email: pens@pens.ac.id

SURAT TUGAS

No. 3786/PL14/TU/2017

Sehubungan dengan pelaksanaan Program Beasiswa Retooling Kompetensi Vokasi Dosen Pendidikan Tinggi Vokasi dari Direktorat Pengembangan Kelembagaan Perguruan Tinggi, saya yang bertanda tangan dibawah ini:

Nama	: Dr. Zainal Arief, S.T., M.T.
NIP.	: 196701281990031002
Pangkat/Gol	: Pembina (IV/a)
Jabatan	: Direktur Politeknik Elektronika Negeri Surabaya/ PENS

Menugaskan kepada nama di bawah ini :

Nama	: Setyo Nugroho, ST, MT
NIP.	: 198611132012121002
Pangkat/Gol	: Penata Muda Tk I (III/b)
Jabatan	: Dosen Sistem Pembangkit Energi

Untuk mengikuti Pelatihan pada bidang Aircraft Technician dengan sertifikasi Internasional Aircraft Technician program Retooling Dosen Vokasi Ristekdikti pada tanggal 5 November 2017-15 Desember 2017 bertempat di ZbW St. Gallen School Swiss.

Direktorat Pengembangan Kelembagaan Perguruan Tinggi menyediakan biaya Pelatihan dan Akomodasi sedangkan biaya lainnya ditanggung oleh Politeknik Elektronika Negeri Surabaya.

Demikian surat tugas ini dibuat untuk dilaksanakan dengan penuh tanggung jawab.

Surabaya, 12 September 2017
Direktur,

Dr. Zainal Arief, S.T., M.T.
NIP. 196701281990031002



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No. : 3787/PL14/TU/2017

12 September 2017

Kepada Yth.
Sekretaris
Direktorat Jenderal Sumber Daya IPTEK dan DIKTI
Kementerian Riset, Teknologi dan Pendidikan Tinggi
Gedung Dikti Lantai 5
Jalan Jenderal Sudirman Pintu I Senayan
Jakarta 10270

Perihal : Permohonan Ijin ke Luar Negeri

Dengan ini kami mohon bantuan untuk memberikan persetujuan ijin tugas dinas ke luar negeri dalam rangka mengikuti kegiatan beasiswa *retooling* kompetensi vokasi dosen pendidikan tinggi vokasi luar negeri ke negara *Swiss* yang akan dilaksanakan tanggal 5 November s.d 15 Desember 2017 atas nama berikut ini:

No.	Nama	NIP	Jabatan
1	Setyo Nugroho, ST, MT	198611132012121002	Dosen Sistem Pembangkit Energi
2			
3			

Biaya yang diperlukan dalam kegiatan ini ditanggung melalui mekanisme DIPA Direktorat Jenderal Kelembagaan Iptek Dikti. Kelengkapan dokumen pendukung lainnya, kami lampirkan bersama surat ini.

Atas perhatian dan kerjasamanya kami ucapkan terima kasih.



Dr. Setyo Nugroho, S.T., M.T.
NIP. 196701281990031002

Tembusan :

1. Menteri Riset, Teknologi dan DIKTI (sebagai laporan)
2. Sekretariat Negara Bidang KTLN

2.Fotocopy paspor:





3. Dokumen output kegiatan, antara lain:

From knowledge to action

Transfer booklet for

Setyo Nugroho

for SVEB Module 1: Training course in

Authors	Harald Daniel Felder	Graschi
Translation	Ashley Slapp	
Version	1.0/ 10.11.2017	

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3. Template for the method evaluation.....	Error! Bookmark not defined.

1. Transfer booklet

What is a transfer booklet?

Many vocational training courses (care professions, metalworking industry, commercial sector) throughout Germany, Austria and Switzerland expect the learners to keep a logbook, learning journal or work log. They represent approved documentation of the daily working processes and are used to reflect on the work and learning results.

At the start of this training course, you stepped into the role of the learner. Any exchange with module instructors and your colleagues within the course group offers the opportunity to further expand your existing competences and question previous practices. This learning environment leaves impressions and stimuli at the moment of exchange which quite often fade (too) fast.

Therefore, you should use this transfer booklet during the training course to develop implementation ideas for your training situation. Within this context, you reflect on your experiences, emotions and thoughts which, based on specific intentions, become visible in your professional educational practice through action. The transfer booklet is structured as follows for each part module:

- Preparatory task
- Transfer task and reflective questions
- Checklist

The entries and documents form the transfer booklet Module 1.

At the **start of the practical demonstrations**, you hand in a copy of the labelled transfer booklet to the course instructor. You keep the original one.

Please only submit the documents requested in the checklist!

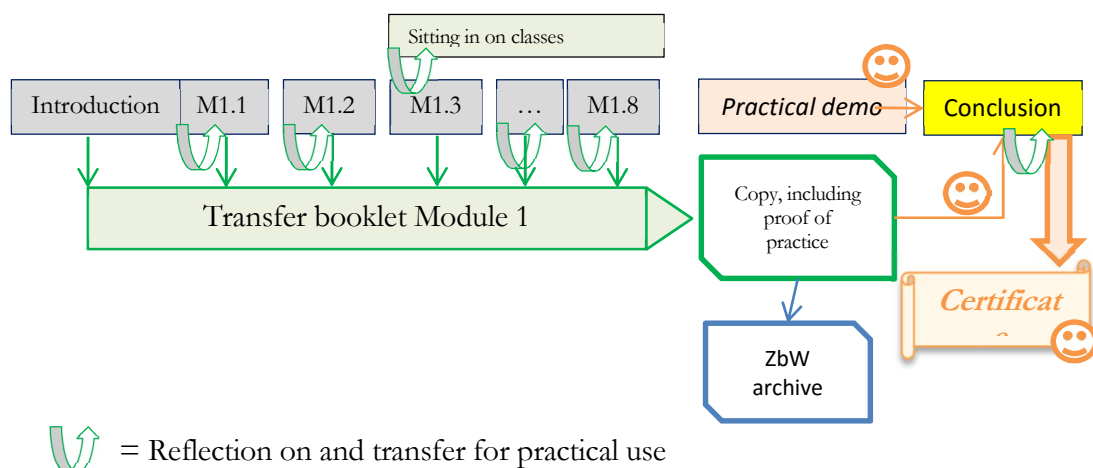


Figure 1: Keeping the transfer booklet

Your transfer booklet is checked by the course instructor for **completeness** according to the checklist and **achievement of the minimum competence**. You receive short, written notes as feedback.

Understanding learning and teaching

Transfer task for the entire training

Read the script for this part module using your own reading strategy.

During this training course, you will be confronted with a vast amount of new technical terms and methods. On the learning platform you will find two glossaries in the course SFAL FA-M1 “Lecturer”.

- Glossary of new words and didactic terms
- Glossary of methods

Organise cooperation within the class in order to fill these two glossaries with content and to comment and improve them.

The “Glossary of new words and didactic terms” is used as a translation of the English terms into your national language Bahasa Indonesian. Add an entry:

- **Term** in English
- **Definition** in Bahasa Indonesian
- **Category** Select the title of the respective script
- **Appendix** Helpful documents for the term can be uploaded here (voluntary)

During the lessons, you will repeatedly encounter and experience applied methods that are discussed together. Use the “Glossary of methods” in the class to document these methods.

Please describe the method or learning strategy in English. Use the following structure:

1. name
2. procedure and social forms
3. time needed
4. purpose
5. effect in class
6. threats
7. variations

»» Reflective questions

Summarise your thoughts in writing using the following questions:

a) Content-related examination

- What have you newly learned? (Knowledge, findings, experiences)
I learned about learning style developed by David Kolb which divides four quadrants of style i.e accommodator, converger, diverger, assimilator
- Which questions regarding the content remain unanswered?
What kind of learning subject that this method most suitable?

b) Professional practice-related examination

- Which differences have you noticed between Switzerland and Indonesia?

About educational system and apprentice

- Formulate at least two goals that you wish to implement in your learning events.
To map the class learning style, to develop teaching style that suitable for each subject

Checklist for this part module

- ☐ Glossary is filled in on the learning platform, commented and improved
- ☐ Reflection

Presenting learning contents

Preparatory task

Part module 1.2 focusses on your presentation in front of the course group and on the preparation of the respective media. In order to enable practical learning and experimenting, prepare yourself according to the following tasks:

1. Read the script for this part module using your own reading strategy.
2. Collect information (texts, images, graphics) for a short presentation lasting roughly 7 to 9 minutes. The content is taken from your professional specialist field or from one of your training sequences. Think about which content you want to present in this time. You can practice something familiar or consciously try out something new. You will hold the presentation **in the last course block of this part module** in front of your colleagues.

Bring along these **not yet prepared** materials/documents and information to the lesson.

Transfer task

As part of the classroom event, you have held a presentation and received feedback.

- Summarise the key aspects of the class feedback.
[Attention to all learners, make a good preparation, manage the time](#)
- What conclusions can you draw from this?
[A good presentation need a good preparation and arrangement](#)
- Describe your adaptations.
[Adapt to learning environment and culture](#)

»» Reflective questions

a) Content-related examination

- What have you newly learned? (Knowledge, findings, experiences)
[Didactic triangle, didactic analysis](#)
- Which questions regarding the content remain unanswered?

[Nothing at this time](#)

b) Professional practice-related examination

- Which differences have you noticed between Switzerland and Indonesia?
[Nothing at this time](#)
- Formulate at least two goals that you wish to implement in your learning events.
[To implement didactic analysis for planning, to determine the competences for each subjects](#)

Checklist for this part module

- ☐ Feedback summary and adaptations
- ☐ Reflection

Planning learning processes**Preparatory task**

1. Read the script for this part module using your own reading strategy. Use the glossary.
2. Bring along the curriculum of your course (if specifications are available) to the classroom event.
3. Bring along (if available) the fine planning of an existing lesson sequence.

Transfer task

Create a complete rough planning (didactic analysis) for a training block consisting of at least 4 units. This includes:

1. Clarifying the resources
2. Analysing and reducing the subject matter
3. Determining the competences, abilities, attitudes to be achieved
4. Formulating the verifiable learning or performance goals
5. Specifying the learning and teaching methods for achieving the formulated goals

[File already uploaded](#)

»» Reflective questions

Summarise your thoughts in writing using the following questions:

a) Content-related examination

- What have you newly learned? (Knowledge, findings, experiences)
[Sandwich model, RITA](#)
- Which questions regarding the content remain unanswered?

Nothing at this time

b) Professional practice-related examination

- Which differences have you noticed between Switzerland and Indonesia?

Nothing at this time

- Formulate at least two goals that you wish to implement in your learning events.
To apply sandwich model of learning, to apply attitude factor at every competence level

Checklist for this part module

- ☐ Rough planning
- ☐ Learning or performance goals, including the specification of the productive methods
- ☐ Reflection

Designing learning environments

Preparatory task

1. Read the entire documentation. Use the glossary.
2. We will develop an exemplary advance organiser in the classroom event. Bring along a learning topic and documents/materials from a lesson or teaching situation which might be suitable for the visualised networking of the topic (AO).

Transfer tasks

Cognitive apprenticeship:

During your training course at SR-TECHNICS, you will become familiar with various forms of instruction.

- Describe how such instruction works.
They made a written instruction and the instructor brief to the learners and give an example
- Note down key statements and questions during the articulation phases.
The learners teach each other and demand the instructor to check their work
- Describe how the reflection phase goes and which questions you use to check and consolidate achievement of the learning goal.
The learners asked about their learning, what about their feeling and understanding regarding the session

»» Reflective questions:

Summarise your thoughts in writing using the following questions:

a) Content-related examination

- What have you newly learned? (Knowledge, findings, experiences)
[Cognitive Apprenticeship](#)
- Which questions regarding the content remain unanswered?
[Nothing at this time](#)

b) Professional practice-related examination

- Which differences have you noticed between Switzerland and Indonesia?
[In Switzerland, scaffolding and articulating step is really emphasized](#)
- Formulate at least two goals that you wish to implement in your learning events.
[To provide a manual procedure as scaffolding step, implement articulation method in learning process](#)

Checklist for this part module

- ☐ Completed advance organiser, planning of a reciprocal teaching and learning method and description of a cognitive apprenticeship
- ☐ Reflection

Instructing group processes**Preparatory task**

Group-dynamic processes are described in chapter «**Instructing group processes**». Group dynamics investigates typical patterns, processes and roles. It deals with the questions of how groups develop under certain conditions and how such development processes can be supported.

1. Read the entire documentation. Use the glossary.
2. Note down the teaching situations where you have already experienced such processes.

Transfer task

1. Create a visualisation of the group (drawing, mind map, network, sociogram).
2. Add words, symbols or signs to the aspects of group dynamics you want to highlight.
Examples:
 - Characteristics of a group
 - Role distribution within the group
 - Relation between you as the trainer and the group
 - Behaviour of group members
3. Describe a specific or possible disruption.
4. Describe a proactive and a reactive strategy for the situation stated above.

The group consist of one leader to moderate all off member ideas. All of ideas will be considered together to make an agreement an commitment. Each member shares all of their ideas, and the teacher should encourage them to think and share anything.

The disruption could occur when there is a member who do not actively participating of the discussion and will be not happily to follow the agreement or commitment

Proactive strategy: explain the topic or the project in detail, make a commitment in advance to do this activity together

Reactive strategy: the leader should motivate and encourage all the members, respect all of the ideas.



»» Reflective questions

Summarise your thoughts in writing using the following questions:

a) Content-related examination

- What have you newly learned? (Knowledge, findings, experiences)
[Motivational type, group process, class management](#)
- Which questions regarding the content remain unanswered?
[Nothing at this time](#)

b) Professional practice-related examination

- Which differences have you noticed between Switzerland and Indonesia?
[More group activities in CH](#)
- Formulate at least two goals that you wish to implement in your learning events.
[Make more group activities, To know about motivational type of the learners](#)

Checklist for this part module

- ☐ Visualisation and description of a group
- ☐ Description of a proactive and a reactive strategy
- ☐ Reflection

Enabling self-controlled learning

Preparatory task

1. Read the documentation for the entire topic. Use the glossary.
2. Review your learning behaviour and the learning processes during the weeks since your arrival in Switzerland.
 - a. How could you learn best and implement the acquired knowledge in your practice?
 - b. What impaired your learning process or the transfer?
 - c. On which levels (action, interaction, reflection) did you receive support and encouragement and from whom?
 - d. Which expectations do you have regarding implementation at your school?

Transfer task

Experiences have shown that “self-controlled learning” can, depending on the respective field of training, be implemented in various ways. Therefore, you can choose between two topics for the transfer task and work on the one that suits your training practice or the level of your participants best.

Variant 1: Learning strategy

- Deal with a learning strategy of your choice (script in chapter 3) in depth.
[I choose a group work for learning using group puzzle method. The method used to discuss and determine what to do in an aircraft to overcome the atmosphere condition](#)
- Describe how you convey this strategy to your learners and use it in your lessons.

I write some material theories and give to the learners. Each of them will read, some will focus on a topic, then I divide them into three groups consisting at least a member is mastered of a topic then they will share the knowledge

»» Reflective questions

Summarise your thoughts in writing using the following questions:

a) Content-related examination

- What have you newly learned? (Knowledge, findings, experiences)
To know better of the group resources and how to manage them
- Which questions regarding the content remain unanswered?
Nothing at this time

b) Professional practice-related examination

- Which differences have you noticed between Switzerland and Indonesia?
In CH more chance to compromise instead of voting
- Formulate at least two goals that you wish to implement in your learning events.
To make a project group with certain topic, To give a more role to students to manage their own group

Checklist for this part module

- ☐ Written working task or implementation of a learning strategy
- ☐ Reflection

Assessing learning performance

Preparatory task

1. Read the documentation for the entire topic. Use the glossary.
2. Bring along your own written examination or your own evaluation grid for an oral/practical examination to the classroom event.

Transfer task and reflection

In the classroom event, you have either revised an examination or an evaluation grid or created a new draft of either of them.

Complete one item of work. The task is completed when you have

Variant 1: newly created written examination

- described the competences or learning goals to be assessed,
- designed an examination or a learning goal assessment in line with the quality criteria
- created a respective sample solution with points allocation and
- determined the minimum competence you expect and a grade or evaluation scale.

Using learning goal oriented methods

Competence

Trainers know various collective and individual teaching methods and use them in line with the respective requirement and situation.

Task

Take your curriculum and a lesson sequence (e.g. four units).

1. Formulate the competence(s) to be developed for this precise lesson sequence.
2. Based on the competence, formulate the learning goals to be achieved.
3. Check the formulated goals together with a colleague and discuss them with regard to
 - clear wording
 - achievability within the lesson sequence
 - verifiability

Note down the target examples on an A4 sheet (observe legibility).

1. With which methods (collective and individual) do you achieve this learning goal? Search for a large number of approaches to achieve the learning goal.
2. How much time do you plan for each method to achieve the specified learning goal? Which methods are suitable for your lesson? Underline them.

C level	Your own learning goal example	Collective method (monologue/dialogue)	Individual method (in the social form IW, PW, GW)	Assessment (oral/written)
C1	To list about fluid properties	Discussion in class	A quiz	Please mention three of fluid properties ! (oral)
C2	To explain about hydrostatic	Discussion in class	Explain as pair discussion	Please explain about hydrostatic! (oral)
C3	To determine the buoyancy force	Give some presentation and demo	Doing some experiment	Given some shape of things, then calculate the buoyancy force and determine whether it is float or not. (written)
C4	To determine the	Give some presentation	Observing the practical	Observe and explain why the

C5	cause of pressure drop	Give a Lecture, give an example	module and group discuss	pressure drops (written)
	To calculate the pump's power		Individual exercise, training teaching reporting	Given a case, calculate the best choice of pump regarding its performance (written)
C6	To design a piping system	Progress meeting periodically	Simulation, stationary work	Give a project to design piping system in a house or small building (project)

»» Reflective question

- Which knowledge have you acquired for designing the examination process or evaluation practice in your own field of training? [How to prevent biases in oral examination, how to justify whether a question is selective.](#)

Checklist for this part module

- ☐ Creation of an examination or an evaluation grid or revision
- ☐ Reflection

Preparatory Materials

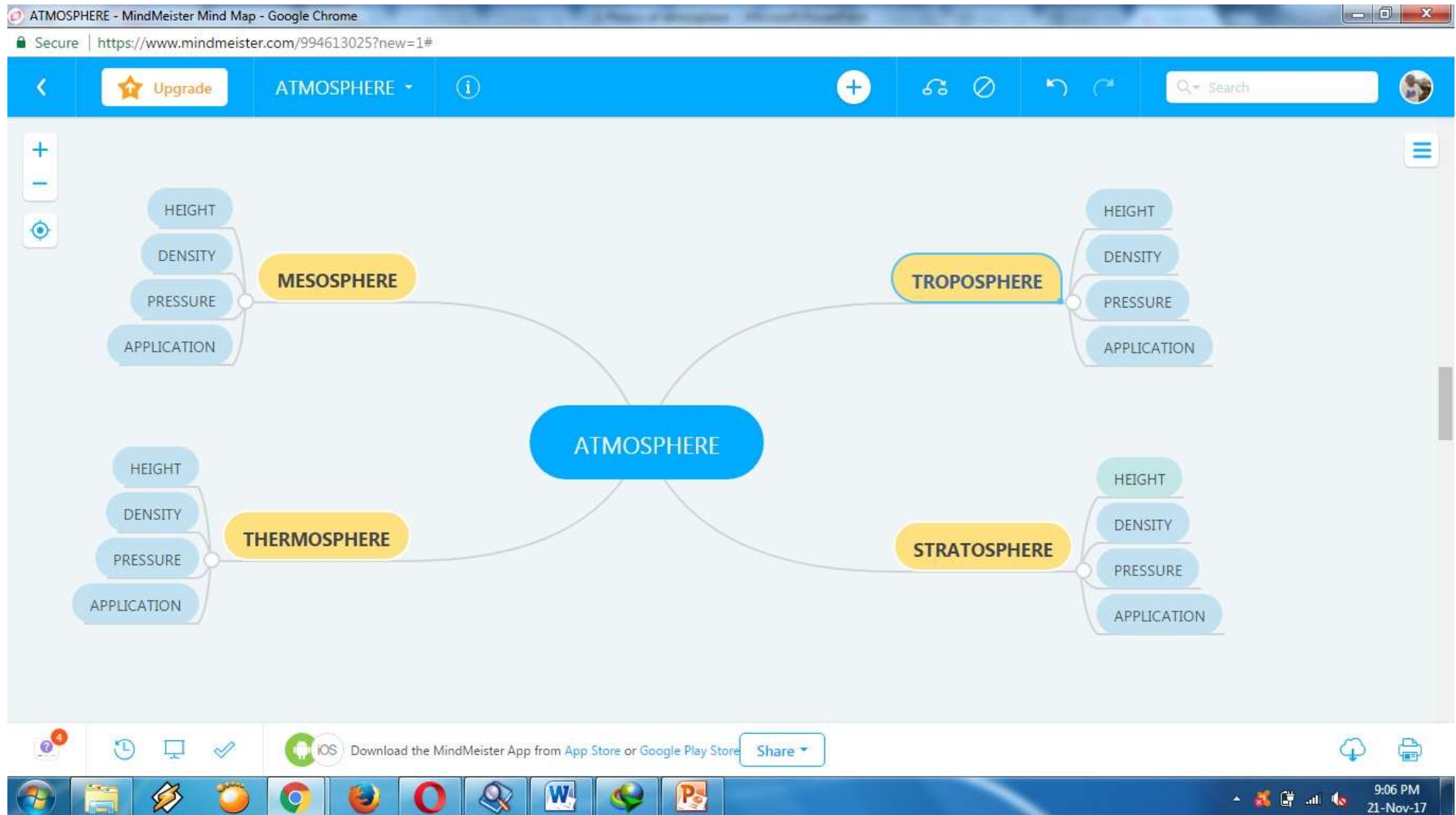
Setyo Nugroho

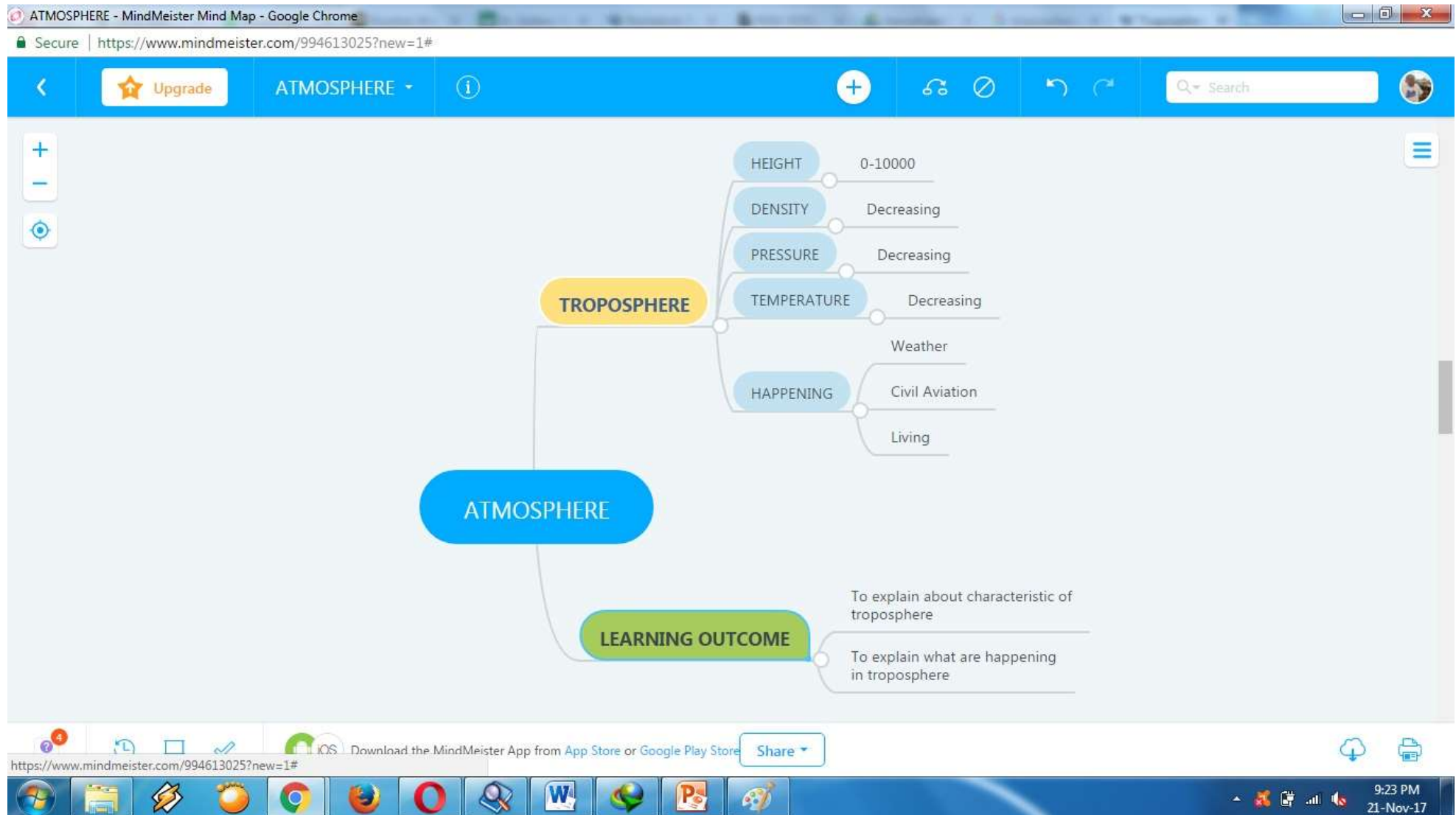
Didactic analysis for a learning event

To be able to plan learning events in a targeted manner, the prerequisites and framework conditions need to be clarified. Didactic analysis is made of an environment analysis and a fact analysis.

Analyze a learning event following the tasks below and the questions of the workbook (planning learning processes) pages 13/14.

Environment analysis	Fact analysis
Environment • To follow the IQF and Airworthiness Regulation • How to compete with foreign workers • Occupy the MRO industry workers demand Educational institution • Vision and Mission of PENS • Airworthiness Regulation • Learning outcomes of study programme • Updated Curriculum • Classroom, Laboratories, Workshop • Internship Programme • Study Excursion • In House Training • Practical modules	Trainers • Update with recent technology • Update with recent learning style and method • Update with recent Airworthiness Regulation • Give a comprehensive understanding about aircraft system • Give problem solving and troubleshooting cases • Give a proper model as aircraft technician Learners • Might be unfamiliar with aircraft environment • Have a technical engineering background • Have a strong purpose and motivation • Demand a lot of examples of science application Content • Basic fundamental of air physics • Concept of technology of aircraft • Applied engineering in an aircraft





Analysing and reducing the subject matter

Analyse the topic and the contents of your learning event, regarding the prerequisites. Then reduce the content from the four angles below.

Angle 1: Knowledge	Angle 2: Apply in practice	Angle 3: My view as a teacher	Angle 4: View of the learners
What is important for the learners to know and/or to prove on an exam?	What should the learners be able to apply in practice?	What are my focal points as an expert and teacher?	What might be interesting for the learners?
To know the air characteristic in troposphere To know what to do to make aircraft endure the atmosphere condition	To determine the tendency of air properties in atmosphere	Know how the air properties vary in the atmosphere Know the system in an aircraft to endure the atmosphere condition	Air properties vary with in the atmosphere There are many system in an aircraft to endure the atmosphere condition

Using learning goal oriented methods

Competence

Trainers know various collective and individual teaching methods and use them in line with the respective requirement and situation.

Task

Take your curriculum and a lesson sequence (e.g. four units).

4. Formulate the competence(s) to be developed for this precise lesson sequence.
5. Based on the competence, formulate the learning goals to be achieved.
6. Check the formulated goals together with a colleague and discuss them with regard to
 - clear wording
 - achievability within the lesson sequence
 - verifiability

Note down the target examples on an A4 sheet (observe legibility).

3. With which methods (collective and individual) do you achieve this learning goal? Search for a large number of approaches to achieve the learning goal.
4. How much time do you plan for each method to achieve the specified learning goal? Which methods are suitable for your lesson? Underline them.

C level	Your own learning goal example	Collective method (monologue/dialogue)	Individual method (in the social form IW, PW, GW)
C1	To list about air properties	Discussion in class	A quiz
C2	To explain about air properties	Discussion in class	Explain as pair discussion

M1.4 Planning learning processes

C3	To determine the air properties tendency regarding the height in the troposphere	Give some presentation and demo	Doing some experiment and reflected experience
C4	To determine the cause of the tendency of air properties regarding the height in the troposphere	Give some presentation	Group discuss
C5	To determine the air properties at any height in the troposphere	Give a Lecture, give an example	Individual exercise, training teaching reporting
C6	To design a aircraft system to endure the atmosphere condition	Progress meeting periodically	Group work, stationary work

Process planning (script)

Topic/Situation
ATMOSPHERE

Competences

Students could explain about troposphere

Learning goals

After study this topic, students could:

1. Explain about troposphere characteristic
2. Explain about what are happening in troposphere
3. To know what to do on an aircraft to endure the atmosphere condition

Time	Teaching and learning steps (content/goal/organisation/information/key questions)	Social forms and methods	Media material and
0	Opening, Explain the overview and learning	Collective	Flipchart, Whiteboard
5	outcome	IW, Plenum	Environment
10	Students observe the atmosphere	Collective	Pinboard
20	Explain about troposphere	GW	Flipchart
35	Students do Group Puzzle	PW, Plenum	Flipchart, Whiteboard
40	Conclude and reflection	Collective	-
	Closing		

Abbreviations:

lecture, demonstration
 Individual:

Collective: M = monologue,
 D = dialogue, discussion, presentation
 IW= individual work, PW= partner work, GW= group work

ATMOSPHERE PHYSICAL CHARACTERISTIC

This subject will discuss about how the physical characteristic in atmosphere, especially in troposphere layer. And how the aircraft engineers design systems to handle about this characteristic.

Learning Goals

1. To know the properties of air in troposphere
2. To determine the properties of air regarding the altitude
3. To know what to do to make an safe flight which operates in troposphere

Air Properties

Temperature

The temperature of the troposphere generally decreases as altitude increases. The reason for this temperature difference is that the ground absorbs most of the sun's energy, which then heats the lower levels of the atmosphere with which it is in contact. Meanwhile, the radiation of heat at the top of the atmosphere results in the cooling of that part of the atmosphere

Pressure

The pressure of the atmosphere is maximum at sea level and decreases with altitude. This is because the atmosphere is very nearly in hydrostatic equilibrium so that the pressure is equal to the weight of air above a given point.

Density

Air density, like air pressure, decreases with increasing altitude. It also changes with variation in temperature and humidity. The reason is the nearer to the ground the more the gravity will attract the air molecules.

Table 1. Standard Air Properties

H (km)	T (°C)	P (kPa)	ρ (kg m ⁻³)
-1	21.5	113.920	1.3470
0	15.0	101.325	1.2250
1	8.5	89.874	1.1116
2	2.0	79.495	1.0065
3	-4.5	70.108	0.9091
4	-11.0	61.640	0.8191
5	-17.5	54.019	0.7361
6	-24.0	47.181	0.6597
7	-30.5	41.060	0.5895
8	-37.0	35.599	0.5252
9	-43.5	30.742	0.4664
10	-50.0	26.436	0.4127
11	-56.5	22.632	0.3639
13	-56.5	16.510	0.2655
15	-56.5	12.044	0.1937

4. Surat-surat pendukung dari Perguruan Teknik di Luar Negeri



Embassy of Switzerland
Jl. H.R. Rasuna Said Blok X 3/2
Kuningan, Jakarta 12950
Indonesia

and
To whom it may concern

Jakarta, October 27th, 2017/CvH

Invitation Letter

Dear Madam/Sir

We herewith would like to invite the instructors (name list enclosed) of Indonesian Polytechnics as Delegation from the Ministry of Technology, Research and Higher Education to participate between November 13th and December 22nd, 2017 in a Vocational Education Training (VET) at the Zentrum für berufliche Weiterbildung (ZbW), St Gallen, Switzerland.

The Training Program is jointly organized by the Ministry of Technology, Research and Higher Education and SITECO.

During their stay in Switzerland, accommodation is organized by ZbW in coordination with SITECO at the: Herberge Rorschach, Churerstrasse 4, CH-9400 Rorschach.

This invitation letter serves the purpose of Schengen multiple entry visa application.

The Ministry of Technology, Research and Higher Education in cooperation with SITECO guarantees costs and expenses during their stay in Switzerland and takes over responsibility for their repatriation after the visit.

Thank you for your support, and best regards,

A handwritten signature in black ink, appearing to read "Christine von Harrach".

Christine von Harrach

SITECO Association for Swiss International Technical Connection

List of Participants

No	Name	Passport Number	Valid till	Representative of Institution	Location
1	Mochamad Luthfi	B7985241	September 12 th , 2022	Politeknik Negeri Bandung	zbw Zentrum für berufliche Weiterbildung St. Gallen
2	Tria Mariz Arief	B1789875	September 7 th , 2020	Politeknik Negeri Bandung	zbw Zentrum für berufliche Weiterbildung St. Gallen
3	Wowo Rossbandrio	B7390496	July 13 th , 2022	Politeknik Negeri Batam	zbw Zentrum für berufliche Weiterbildung St. Gallen
4	Didi Istardi	B1536203	August 24 th , 2020	Politeknik Negeri Batam	zbw Zentrum für berufliche Weiterbildung St. Gallen
5	Salamat Muslimin	A6426428	September, 12 th , 2018	Politeknik Negeri Sriwijaya	zbw Zentrum für berufliche Weiterbildung St. Gallen
6	Muhamad Rifai	B3697212	April 6 th , 2021	Politeknik Negeri Malang	zbw Zentrum für berufliche Weiterbildung St. Gallen
7	Ariyadi	B4314621	July 28 th , 2021	Politeknik Negeri Sriwijaya	zbw Zentrum für berufliche Weiterbildung St. Gallen
8	Martinus Mujur Rose	B8012228	September 8 th , 2022	Politeknik Negeri Sriwijaya	zbw Zentrum für berufliche Weiterbildung St. Gallen
9	Ari Wibowo	B0115813	January 14 th , 2020	Politeknik Negeri Batam	zbw Zentrum für berufliche Weiterbildung St. Gallen
10	Setyo Nugroho	B7623291	September 13 th , 2022	PENS	zbw Zentrum für berufliche Weiterbildung St. Gallen
11	Denda Dewatama	B5173729	October 17 th , 2021	Politeknik Negeri Malang	zbw Zentrum für berufliche Weiterbildung St. Gallen
12	Syariful Hikmah Sornin	B5888807	December, 4 th , 2021	Politeknik Negeri Medan	zbw Zentrum für berufliche Weiterbildung St. Gallen
13	Jhony Hartanta Sembiring	A6769889	November 1 st , 2018	Politeknik Negeri Medan	zbw Zentrum für berufliche Weiterbildung St. Gallen
14	Arif Febriansyah Juwita	B0813132	April 06 th , 2020	Politeknik Negeri Batam	zbw Zentrum für berufliche Weiterbildung St. Gallen
15	Soni Hestukoro	B0371380	February 5 th , 2020	Politeknik Negeri Medan	zbw Zentrum für berufliche Weiterbildung St. Gallen

5. Fotocopy Sertifikats

SFAL Lecturer



Practical Demonstration Lesson SFAL Lecturer (FA-M1 SVEB)

Name: Setyo Nugroho

St. Gallen 1711

Subject: Atmosfer

Proof of competences

passed:



not passed:



Place / Date: St. Gallen, December 19th 2017

Expert:

Co-Expert

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page 1

SFAL Lecturer



Per subskill, each criterion is evaluated with 2 points (met), 1 point (partially met) or 0 points (not met). The indicators in the central column are used as guidance and do not have to be fully observed to achieve the maximum number of points. A subskill is met when achieving four or more points.

Didactic design: Quality of the preparatory materials		Strengths / Potential for development / Tips
Planning	Prerequisites (didactic triangle) have been clarified. ☒ The functional structure of the subject matter can be observed. ☒ The weighting and reduction of the subject matter is visible. ☒ The learning goals describe the content and observable behaviour. ☒ The lesson planning is comprehensible (time, method, media, etc.) ☒ The learning goals have been adapted to the conditions and the addressees. ☒ The examples and tasks are relevant for the training practice. ☒ The learning goals can be achieved using the applied methods. ☒ The structure is well designed and a recurring theme can be identified. ☒ The documents (worksheets, scripts, handouts) are complete.	• Didactic triangle has been clarified for the own school, not for this class • Structure of the subject matter is shown • Excellent assignment for the group puzzle with thematic reference
Consistency	☒ Media are reduced, visualised, legible and appealing. ☒ Various media are used in an adapted manner. ☒ Sources are specified in the teaching materials.	
Design		
Methodological design: Supporting an active learning process		Strengths / Potential for development / Tips
Architecture	Frontal instruction teaching style Interrupted frontal instruction teaching style (collective > individual) Sandwich (collective > individual)	
Structure	☒ Designs an addressee-appropriate introduction into the teaching unit. ☒ Illustrates the structure and the goals at the right places. ☒ Selects various forms of interaction (at least three). ☒ Concludes the topic in terms of content and organisation. ☒ Selects different and targeted methods. ☒ Promotes key action competences of the topic. ☒ Achieves the learning goals or at least initiates the transfer. ☒ Applies methods, including junctions, appropriately. ☒ Takes the necessary resources (knowledge, experiences, abilities, attitudes) into account for developing competences. ☒ Observes the entire learning process (e.g. RITA). ☒ Enables independent learning of the participants. ☒ Offers pre-structured aids for learning and remembering. ☒ Adheres to the time structure.	• Humorous introduction to the subject matter. • After the main information the group work brings the possibility to go further into the subject matter • Clear organization of the work and who is the leader • Discussion about pressure, density and temperature could be commented or asked a deepening question • overall excellent lesson structure, where the goals are achieved and the methods make sense
Consistency		
Design		

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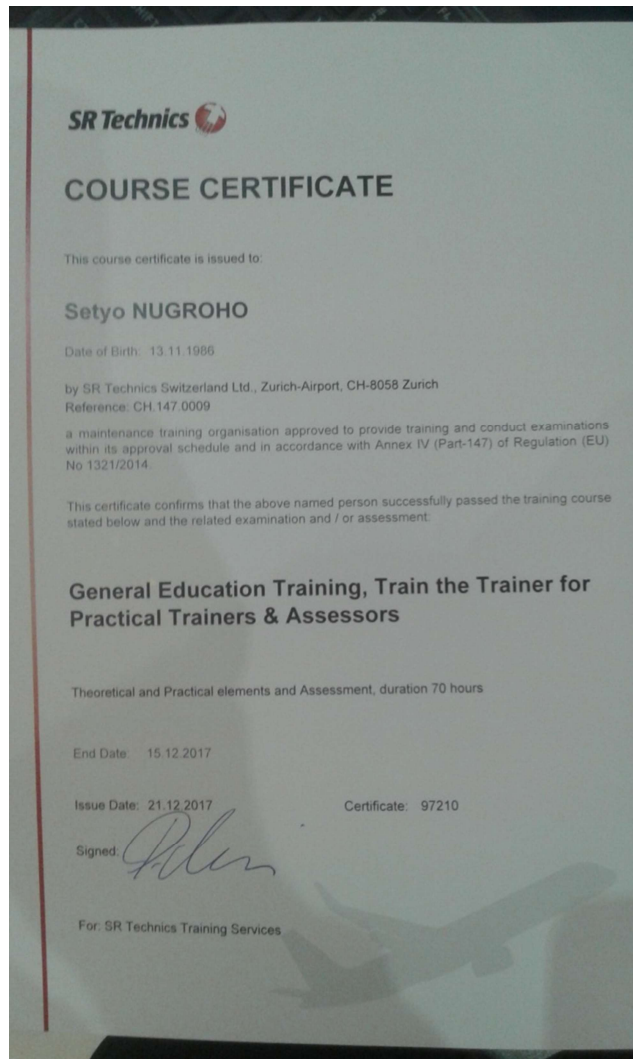
SFAL Lecturer



Personal competence: Linguistic expression, appearance, effect		Strengths / Potential for development / Tips
Language 2 1 0	✓ Uses linguistic styles and conveys contents in a vivid manner. ✓ Repeats core statements and introduces technical/key terms. ✓ Speaks clearly and uses coherent sentences. ✓ Expresses himself/herself in a precise and comprehensible manner. ✓ Varies the speech rate/pitch. ✓ Speaks freely. ✓ Comes across as being credible/committed/authentic. ✓ Uses facial expressions and gestures appropriately and naturally. ✓ Keeps eye contact with everyone, addresses all the learners. ✓ Is able to respond flexibly to unforeseeable events. ✓ Is competent in the operation of devices and technology. ✓ Designs media in a legible, reduced, multi-coded and appealing manner.	• Terms are explained step by step • Structured presentation at the flipchart with a goal-oriented visualization • Speaks freely, clearly and coherent • Appears confident and convincing • Very secure also in detailed explanations • The audience listen to the lecturer and is attentive
Appearance 2 1 0		
Media 2 1 0		
Social competence: Guidance and presentation, perception of the group		Strengths / Potential for development / Tips
Guidance 2 1 0	✓ Guides the learners through a structured lesson. ✓ Keeps an eye on the objective. ✓ Structures/guides the discussion. ✓ Gives stimuli/asks open questions. ✓ Controls the comprehension of his/her own expressions. ✓ Allows sufficient time for thinking and answering questions. ✓ Establishes connections between the participants' answers and contents. ✓ Provides encouraging and resource-based feedback. ✓ Addresses specific needs in the individual phases. ✓ Acts appropriately and in a solution-oriented manner in the event of disruptions. ✓ Treats objections in a matter-of-factly fashion.	• Excellent questions to the class – stimulates attendance and thinking • Can answer the questions fact-oriented and patiently. If he doesn't know the answer exactly, he makes it transparent. • There is always a risk to let the participants ask open-minded – you react professionally and with a vivid and humorous spirit
Presentation 2 1 0		
Group 2 1 0		
Capacity for reflecting based on the preparation and the practical demonstration		Strengths / Potential for development / Tips
Preparation 2 1 0	✓ Assesses the quality of the planning and preparation realistically. ✓ Analyses the learning goal formulation and achievement appropriately. ✓ Describes the practical demonstration by detecting good and difficult situations. ✓ Is aware of his/her role and effect in the class. ✓ When asked, assesses individual situations correctly. ✓ Names precise conclusions for these situations and develops solution-oriented action alternatives. ✓ Draws correct conclusions.	• Can reflect in every sense and level • Consistent self-assessment and external evaluation
Execution 2 1 0		
Action alternatives 2 1 0		

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Zentrum für berufliche
Weiterbildung

Gaisenswaldstrasse 6
9015 St. Gallen

Certificate

Certified Trainer SFAL (AdA FA-M1) „Teaching training courses for adults“

Setyo Nugroho, 1986, from Indonesia

has attended and successfully completed the course for the acquisition of the
Module 1 Certificate from

13. November 2017 – 21. December 2017

at Zentrum für berufliche Weiterbildung St. Gallen (Centre for Higher Vocational Education
St. Gallen)

Contents:

- Basic principles of adult-specific teaching and learning
- Formulation of objectives and selection of contents
- Fundamental principles of communication
- Preliminary considerations when planning courses, simple didactic models, lesson planning
- Methods and social settings in adult education
- Basic principles of group dynamics
- Diversity in learning groups
- Simple methods of checking the attainment of objectives and of evaluation
- Adapting learning materials

Conditions for the attainment of the certificate are the completion of the learning objective controls, the passing of an evaluated practical demonstration and a minimum experience in adult education of 2 years and 150 hours.

Module 1 corresponds to 13.5 ECTS points, 405 learning hours

St. Gallen, 21. December 2017

ZbW Zentrum für berufliche Weiterbildung

Dr. Andreas Schubiger
Director

Dr. Gregor Thurnherr
Head of Departement

This module certificate is unlimitedly recognized by the Swiss Federation for Adult Learning (Schweizerischer Verband für Weiterbildung SVEB) and the Committee of Quality Assurance CQA for the admission to the central verification for the acquisition of the Federal Diploma of Higher Education as Adult Educator. The duration, learning objectives, contents and proofs of competence comply with the specifications and guidelines of the CQA. Recognition number: SG141228-M1_AKV-296_1711-9

AdA ■ SVEB / FSEA
FFA ■ SVBA / FSFE / FSFA
SIGEB / CIFAS

6. Curriculum Vitae



Departemen Teknik Mekanika dan Energi
Politeknik Elektronika Negeri Surabaya

DAFTAR RIWAYAT HIDUP

IDENTITAS DIRI

Nama	: Setyo Nugroho
NIP/NIK	: 1986111320121002
NIDN	: 0013118601
Tempat & Tanggal Lahir	: Jakarta, 13 November 1986
Jenis Kelamin (L/P)	: Laki-laki
Status Perkawinan	: Menikah
Agama	: Islam
Golongan / Pangkat	: III/b
Jabatan Akademik	: -
Perguruan Tinggi	: Politeknik Elektronika Negeri Surabaya
Alamat	: Jl. Raya ITS Surabaya
Telp./Faks	: 0315947280/0315946114
Alamat Rumah	: Jl Semampir Barat 8/18 Surabaya
Telp./HP/Faks	: 08562124670
Alamat e-mail	: setyo@pens.ac.id



RIWAYAT PENDIDIKAN PERGURUAN TINGGI

Tahun Lulus	Program Pendidikan	Perguruan Tinggi	Jurusan/Program Studi
2010	Magister	Institut Teknologi Bandung	Teknik Aeronotika dan Astronotika
2008	Sarjana	Institut Teknologi Bandung	Teknik Aeronotika dan Astronotika

PELATIHAN PROFESIONAL*

Tahun	Jenis Pelatihan(Dalam/ Luar Negeri)	Penyelenggara	Jangka Waktu
2017	Teknik Refrigerasi (Dalam Negeri)	PT. MMS	7-10 Agustus 2017
2017	Pelatihan Asesor Kompetensi BNSP Bidang Elektronika (Dalam Negeri)	ITS	Januari 2017
2014	Autodesk Inventor Hands On Workshop (Dalam Negeri)	Autodesk	Oktober 2017
2014	Process Flow Diagram (Dalam Negeri)	PENS	Nov-17
2014	Piping & Instrumentation Diagram (Dalam Negeri)	PENS	Nov-17
2014	Single Cylinder Engines 2.2 KW (Dalam Negeri)	GUNT Hamburg	Februari 2014
2014	Comparison of Pumps (Dalam Negeri)	GUNT Hamburg	Februari 2014
2014	Teaching & Simulation CAN Bus Control & Logger (Dalam Negeri)	Graphtec	Apr-14
2013	Penggunaan MATLAB Simulink untuk Signal Processing dan Sistem Kendali Cerdas (Dalam Negeri)	PENS	Maret 2013
2013	Generic Coal Fired Power Plant (Dalam Negeri)	PT. IPMOMI	Januari 2013
2013	Introduction to ANSYS FLUENT (Luar Negeri)	CAD-IT	Januari 2013

PENGALAMAN MENGAJAR			
Mata Kuliah	Program Pendidikan	Institusi/Jurusan/ Program Studi	Tahun Akademik
Mekanika Fluida 1	D4	Sistem Pembangkitan Energi PENS	2013-2017
Mekanika Fluida 2	D4	Sistem Pembangkitan Energi PENS	2013-2017
Praktikum Mekanika Fluida 1	D4	Sistem Pembangkitan Energi PENS	2013-2017
Praktikum Mekanika Fluida 2	D4	Sistem Pembangkitan Energi PENS	2013-2017
Perpindahan Panas 1	D4	Sistem Pembangkitan Energi PENS	2012-2013
Perpindahan Panas 2	D4	Sistem Pembangkitan Energi PENS	2013-2014
Praktikum Perpindahan Panas 1	D4	Sistem Pembangkitan Energi PENS	2012-2013
Praktikum Perpindahan Panas 2	D4	Sistem Pembangkitan Energi PENS	2013-2014
Mekanika Teknik	D4	Sistem Pembangkitan Energi PENS	2014-2015
Ilmu Material	D4	Sistem Pembangkitan Energi PENS	2014-2015
Komputasi Dinamika Fluida	D4	Sistem Pembangkitan Energi PENS	2013-2017
Praktikum Komputasi Dinamika Fluida	D4	Sistem Pembangkitan Energi PENS	2013-2017

PRODUK BAHAN AJAR*			
Mata Kuliah	Program Pendidikan	Jenis Bahan Ajar	Sem/ Tahun Ajaran
Praktikum Mekanika Fluida 1	D4	Cetak	Sem 1 / 2014-2015
Praktikum Mekanika Fluida 2	D4	Cetak	Sem 2 / 2015-2016

PENGALAMAN PENELITIAN*				
No	Tahun	Judul Penelitian		
			Ketua/ Anggota	Sumber Dana
1.	2016	Uji performansi guide Vanes pada penampang rectangular elbow 90 ⁰ untuk meningkatkan efisiensi sistem perpipaan	Ketua	Lokal
2	2016	Rancang Bangun Turbin Angin Clark-Y Dengan Winglet dan Automatic Pitch Control	Anggota	Ristekdikti
3	2015	Pengaruh Diffuser NACA 14114 2,5R Terhadap Performa Turbin Air Darrieus	Ketua	Lokal
4	2013	Rancang Bangun Terowongan Angin Sirkuit Terbuka Untuk Pengujian Dinamika Fluida	Anggota	Lokal
5	2012	Studi Eksperimental Turbin Angin Tipe Darrieus Untuk Sistem Pembangkit Listrik Tenaga Bayu	Anggota	Lokal
6	2012	Analisis Komputasi Dinamika Fluida Pada Perancangan Turbin Arus Pasang Surut Vertikal Untuk Pembangkit Listrik Energi Terbarukan Arus Laut	Anggota	Lokal

JABATAN DALAM PENGELOLAAN INSTITUSI*

Peran/ Jabatan	Institusi	Tahun
Kepala Laboratorium Fluida	Departemen Teknik Mekanika dan Energi PENS	2013-sekarang

Semua data yang saya isikan dalam biodata ini adalah benar dan dapat dipertanggungjawabkan secara hukum. Apabila di kemudian hari ternyata dijumpai ketidak sesuaian dengan kenyataan, saya sanggup menerima sanksi.

Demikian biodata ini saya buat dengan sebenarnya untuk memenuhi salah satu persyaratan dalam pengajuan sebagai pelamar Beasiswa Retooling Kompetensi Vokasi Dosen Pendidikan Tinggi Vokasi pada Kementerian Riset, Teknologi, dan Pendidikan Tinggi

Surabaya, 14 Agustus 2017

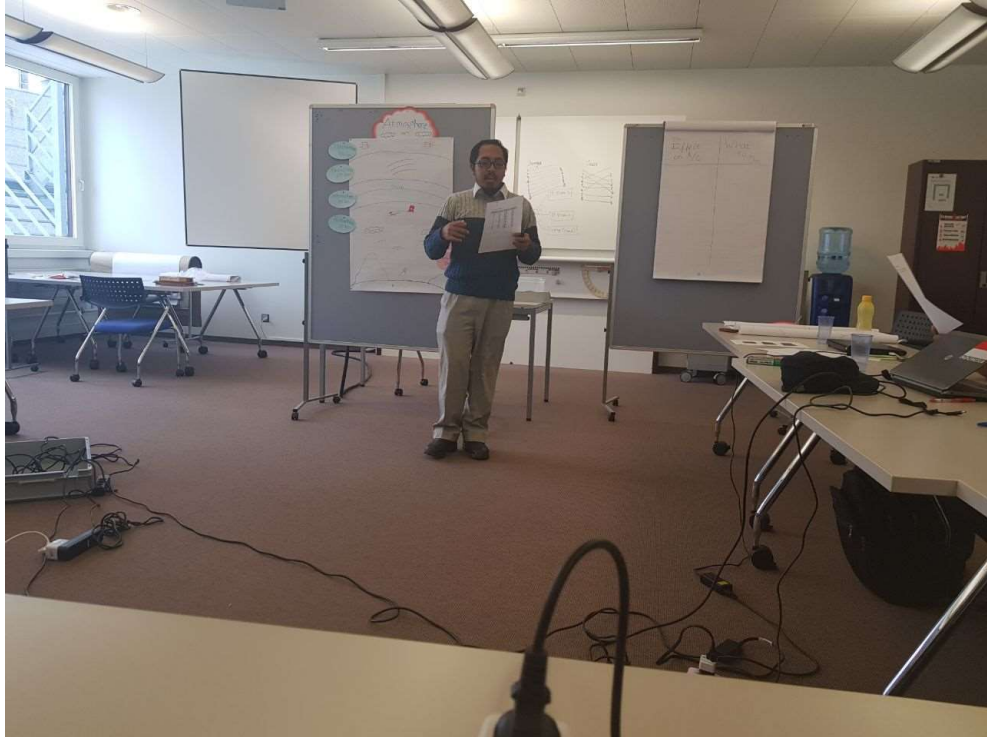
Yang Menyatakan,



Setyo Nugroho

198611132012121002

7. Foto-foto kegiatan (akademik) yang penting dan relevan



Gambar ujian presentasi di ZbW



Modul pembelajaran di SR Technic

